RawTFNet: a lightweight CNN for speech anti-spoofing

Reference notes for privacy-preserving edge deepfake-detection gateway thesis

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Source

Xiao, Y., Dang, T., and Das, R. K. *RawTFNet: A Lightweight CNN Architecture for Speech Anti-spoofing.* 2025 Asia Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC).

1. Motivation

Recent anti-spoofing models (RawNet2, AASIST, Rawformer) reach strong accuracy on ASVspoof benchmarks but require substantial compute, which limits real-time, on-device deployment. RawTFNet targets the same detection task — distinguishing bona fide from spoofed speech — while keeping parameter count and MACs low enough for edge hardware such as a Raspberry Pi 4.

2. Input format

RawTFNet is end-to-end: it consumes the raw audio waveform directly, with no hand-crafted front end (no MFCCs, no spectrogram, no CQT). Per the paper's training protocol:

- Each utterance is cropped or concatenated to a fixed-length segment of approximately 4 seconds (64,000 samples).
- Training uses the ASVspoof 2019 LA training set; evaluation is on ASVspoof 2021 LA and DF.
- RawBoost augmentation (the “series: 1+2+3” combination) is applied to improve robustness to channel and codec variation.

The fixed-length raw waveform tensor is the only input — all feature extraction happens inside the network's own Sinc layer, described next.

3. Architecture overview

RawTFNet has three stages: a Sinc/DWS-SE-Res2Net frontend, a stack of nine Time-Frequency Convolution (TF-Conv) blocks, and a lightweight classifier.

3.1 Frontend

- **Sinc layer** — 70 trainable Sinc (band-pass) filters convert the raw 1D waveform into a 2D feature map of frequency bins × time steps.
- **ResNet block** (32 filters) captures early local patterns.
- **Three DWS SE-Res2Net blocks** (64 filters each) follow. Standard 2D convolutions are replaced with depthwise-separable (DWS) convolutions to cut computation, each block adding a squeeze-and-excitation (SE) module for channel-wise attention.

Output of the frontend is a high-level feature map x_HFM with shape C × F × T (channels × frequency × time).

3.2 Time-Frequency Convolutions (TF-Convs)

This is the paper's main contribution. Instead of a standard convolution that treats the frequency and time axes uniformly, each TF-Conv block processes them separately, then recombines:

- A 1×1 transition convolution adjusts channel count from C to C′.
- A channel shuffle mixes features, then the tensor is split in half along the channel axis into a frequency branch and a time branch.
- **Frequency branch:** 3×1 depthwise convolution → frequency-wise average pooling → 1×1 pointwise convolution.
- **Time branch:** 1×3 depthwise convolution → time-wise average pooling → 1×1 pointwise convolution.
- Each branch's compact 1D summary is broadcast back over the full 2D map and added element-wise to the corresponding split (residual-style).
- The two enhanced branches are concatenated along the channel axis to form the block output.

The model stacks **nine** TF-Conv blocks, with two 2×2 max-pooling layers (stride 2) inserted between them for downsampling. All layers use batch normalization and ReLU.

3.3 Classifier

Global average pooling followed by a single 1D convolution layer outputs a scalar bona fide / spoofed score.

4. Channel width and the “-16” / “-32” naming

A channel-width parameter τ scales the model's capacity, giving RawTFNet-τ. The paper evaluates two variants:

Model	Params	MACs	2021 LA EER	2021 LA min t-DCF	2021 DF EER
RawGT-ST	0.44M	17.4G	10.25%	—	23.26%
RawNet2	17.60M	0.8G	9.50%	—	22.38%
AASIST	0.30M	8.9G	10.51%	0.488	21.07%
SE-Rawformer	0.37M	6.1G	4.98%	0.318	20.26%
Rawformer-L	0.29M	9.2G	6.83%	0.361	—
RawTFNet-16	0.07M	2.9G	4.50%	0.295	18.53%
RawTFNet-32	0.17M	5.4G	5.05%	0.321	16.82%

Comparison of single anti-spoofing systems on ASVspoof 2021 LA and DF evaluation sets (EER = equal error rate, lower is better).

RawTFNet-16 (τ=16) is the smaller variant — 0.07M parameters, the checkpoint format behind a file named `Best_RawTFNet_16.pth`. Despite being smaller than RawTFNet-32, it reaches a *lower* EER on the 2021 LA set (4.50% vs 5.05%), showing that the time/frequency-separated design keeps accuracy high even at reduced width. RawTFNet-32 generalizes slightly better to the harder, codec-compressed DF set (16.82% vs 18.53%).

5. Ablation findings (on RawTFNet-32)

- Removing the frequency branch: EER rises from 5.05% to 6.91%.
- Removing the time branch: EER rises to 7.08%.
- Removing the channel shuffle: EER rises to 5.74%, with no change in model size.

Both branches contribute meaningfully; the shuffle step matters less but is not free to drop.

6. Duration sensitivity

EER is worst on very short clips (0–2s, over 10%), best around 2–4s (matching the training segment length), rises slightly for 4–8s, then drops again for clips over 8s (based on fewer samples). For a real-time telephony gateway, this means detection windows shorter than ~2 seconds should be expected to underperform the paper's headline figures.

7. Why RawTFNet-16 fits an edge-deployment thesis

At 0.07M parameters and 2.9G MACs, RawTFNet-16 is the most lightweight system in the paper's comparison table — smaller than AASIST, RawNet2, and both Rawformer variants — while matching or beating their accuracy on ASVspoof 2021 LA. That combination of footprint and accuracy is what makes it a reasonable choice for confidential inference inside an ARM TrustZone / OP-TEE enclave on a Raspberry Pi 4, where compute and memory budgets are tightly constrained.

Compiled from: Xiao, Dang & Das, RawTFNet (APSIPA ASC 2025).